POLARIS

Spring Research Project: Mentor Rubric

Mentor Name: Ryan Buechele

Mentee Name: Evan Waddell

Project Title: Bell's Theorem in Quantum Magnets

In a short paragraph, explain the project and its learning objectives:

Bell's Theorem is one of the most celebrated results in quantum information theory, showing that quantum systems are able to utilize entanglement to realize measurement correlations which are impossible in a local, classical system. This project will explore this idea in the Quantum Ising model, a simple 1D toy model of magnetism. At the end of this project, the mentee will gain a working understanding of Bell's theorem and the model, and along the way will learn some basic Python programming, elementary linear algebra, and basic quantum mechanics.

Most of the challenging parts of the calculations done here will be pre-programmed, leaving the mentee to simply explore the results as different parameters are varied, and program a few simple functions and loops for collecting, saving, and plotting data. Before getting into the bulk of the work, the first few weeks will be dedicated to some instructional Jupyter notebooks which demonstrate the needed concepts and provide practice problems.

What are the deliverables? (i.e. plot/figure/image/equation mentee will reach)

- Phase diagram plot for the quantum Ising model, showing two distinct phases and the transition between them.
- Plot showing violation of Bell's inequality as a function of the Ising model's parameter.
- Depending on time, we may explore the behavior of these two plots as different parameters are varied.

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Project Calendar: Indicate what the mentee will be expected to do each week. Work days are indicated in the left column. Include estimated time, resources, in/out of class work, and whether it is completed on their own or with you. The project should take 8-10 hours in total.

Dates:	Tasks:
Week 1 (2/12) Intro to Project	Background material: Basics of Python (done), basics of linear algebra in Python (done), basics of quantum mechanics (~3 hours altogether, 2 already completed)
Week 2 (2/19) Outside of Class Work	Background material: Spin-1/2 physics, Pauli Matrices, and Bloch sphere; reading and working through Jupyter notebook with examples and practice problem (~1 hour)
Week 3 (2/26) In class work day	Work through Bell's theorem examples (on own), including IBMQ example for CHSH, discuss all the quantum weirdness (in class together); (~30 minutes out of class + class time)
Week 4 (3/4) Outside of Class Work	Classical Ising model: read about model, play around with provided Monte Carlo simulation (30 minutes)
Spring Break	None - may add some reading/videos here to strengthen concepts from earlier weeks
Week 5 (3/18)	Quantum Ising model: reading ahead and preparing to do calculation next week. (1 hour)
Week 6 (3/25) In Class Work Day	Compute Magnetization in Ising model vs. g and plot; identify phase transition from this plot. (1 hour) Time permitting: finite size scaling with different numbers of sites.
Week 7 (4/1) In Class Work Day	Calculate Bell violation vs. g, plot. (30 min-1 hour)Time permitting: play around with other parameters to see how this effects behavior.
Week 8 (4/8) Poster Work Day	Put together nice plots/figures, organize and assemble poster. (1-2 hours)
Week 9 (4/15) Present!	

Reading List: (Make sure material is level appropriate.)
Jupyter notebooks provided by me outlining the project; Ising model Wiki, Quantum Ising wiki, Bell's Theorem video (MinutePhysics), CHSH Inequality video (Qiskit), Linear Algebra videos (3Blue1Brown)

Resource List: (This should include lab space, instruments, and software. Note that mentees are not allowed to complete research projects that require safety training operation of lab equipment. If you feel this is detrimental to your project, please petition Polaris leadership.)

Laptop computer with JupyterLab and Anaconda Python3 installed (mentee already has this set up and ready to go)

Other Notes:

Evan has taken a programming course in the fall semester and has some background from high school; he has also (willingly) began working on some of the linear algebra basics over the winter break, so we have a bit of a head start on background material.
